

CERTIFICATES FOR PURE CUBIC THUE EQUATIONS: MERLIN–ARTHUR MEMBERSHIP AND THE COMPACT VANISHING PROBLEM

THE DIOPHANTINE-FABLE EXPEDITION

ABSTRACT. For the pure cubic Thue equation $x^3 - dy^3 = m$, the companion paper gave a deterministic $2^{O(s)}$ decision algorithm. Here we ask what *certifies* solvability. The certificate collapses to a single object: a Thiel-format compact representation of the solution element $\beta = x - y\alpha$ itself. Integrality and the norm equation $N(\beta) = m$ verify deterministically in polynomial time — the latter by multiplicativity of norms on power products, the mechanism behind Lagarias’s Pell certificates and Ge’s equality tests. Membership in **MA** follows unconditionally. Membership in **NP** is equivalent, modulo this architecture, to one newly isolated problem — the **Compact Vanishing Problem** (CVP): decide deterministically whether a fixed coordinate functional vanishes at a compactly represented algebraic integer. We classify CVP into the sum-of-square-roots family, prove that ideal-theoretic and radical-rigidity methods cannot express it, and explain by a vacuity remark why Lagarias’s degree-2 certificates never met it.

1. INTRODUCTION

1.1. From decision to certification. The companion paper [companion] proved that pure cubic Thue equations $x^3 - dy^3 = m$ ($d \geq 2$ cube-free, not a cube; $m \neq 0$; input size $s = O(\log d + \log |m|)$ in binary) are decided, with full solution lists, by a deterministic unconditional algorithm in time $2^{O(s)}$ — a first worst-case bound of this shape for a Thue stratum of Smale’s fifth problem [Sma98]. Its Remark 7.5 named the three obstructions to *polynomial-time* decision: the unit (the principal ideal problem, classically believed hard, polynomial only quantumly [Hal07]); the zero test (certified zero-testing against $2^{O(s)}$ -digit height bounds); and the window (quasi-linear in $R = 2^{\Theta(s)}$ ([companion, Lemma C], after Sha [Sha19]) unless the strong window conjecture is proven).

The natural target below polynomial time is therefore **NP membership**: poly-size certificates, deterministic poly-time verification. This paper is that

Date: Sequel draft of 2026-07-22, assembled from the certificate notes; a companion to the exponential-time paper [companion], whose notation it reuses. Every honesty flag of the notes is carried forward: sketches are labeled, quoted-standard facts are flagged, the status ledger is reproduced in Section 7, and no claim below exceeds what the underlying notes established.

Draft — authorship placeholder.

pivot, and its outcome is asymmetric: two thirds of the verifier’s work is classical and easy, and the entire remaining difficulty isolates into one new problem.

1.2. One rung down: Lagarias’s Pell certificates. Lagarias [Lag79] proved that solvability of binary quadratic Diophantine equations over $\mathbb{Z}$ is in NP, although the fundamental solutions have exponentially many digits: the certificate is a power-product representation, and both verifiable conditions — integrality and the norm equation — are *multiplicative*, hence compatible with power products. (Negative Pell decision is even in $\text{NP} \cap \text{coNP}$ by the same technology, with no classical polynomial algorithm known.) This paper asks what happens to that magic one rung up, at degree 3.

1.3. The two main results, and the crux they isolate. The certificate architecture (Section 2) simplifies an outside reviewer’s proposed three-part scheme (unit + reduced representative + orbit index): all three collapse into the single object β , the solution element itself in compact representation. Verification splits into three steps — integrality, norm, Thue shape — and:

- **Integrality and norm verify deterministically in polynomial time** (Section 2), by the format’s denominator bookkeeping and by norm multiplicativity on power products, the latter a special case of Ge’s deterministic multiplicative-relation testing [Ge93].
- **Proposition 3.1 (MA membership, unconditional).** Pure cubic Thue solvability is in MA, with one-sided error: the remaining condition — the vanishing of one coordinate — is testable modulo random primes, with the prime range calibrated against an explicit height bound extracted from the compact data.
- **The crux (Section 4).** The single non-derandomized step is named: **CVP, the Compact Vanishing Problem** — decide deterministically whether a fixed $\mathbb{Q}$ -linear coordinate functional vanishes at a compactly represented algebraic integer. Cubic Thue $\in \text{NP}$ **iff-modulo** CVP (Corollary 4.1, with the honest gloss given there); conditionally, under derandomization hypotheses implying $\text{MA} = \text{NP}$, membership in NP holds today.
- **The vacuity remark (Remark 4.5).** At degree 2 the module $\mathbb{Z} + \mathbb{Z}\sqrt{d}$ is *all* of $\mathbb{Z}[\sqrt{d}]$: the shape condition is vacuous and CVP never arises. The coordinate condition is genuinely new at degree 3 — CVP, not the unit computation, is the honest frontier of the NP question.

Section 5 explains why CVP resists the classical toolbox: four attack routes — Thiel/Ge equality testing, Blömer’s radical sums, ideal-theoretic reformulations, the Baker route — are each shown inapplicable, for one unified reason.

1.4. Organization. Section 2 fixes the certificate and verifier; Section 3 proves MA membership; Section 4 states CVP, its three faces, and the vacuity remark; Section 5 is the impossibility ledger; Sections 6–7 collect positive fragments, open problems, and the status ledger.

2. THE CERTIFICATE AND THE VERIFIER

2.1. Notation (from the companion). $d = ab^2 \geq 2$ cube-free, not a cube, a, b coprime squarefree; $\alpha = \sqrt[3]{d}$; $K = \mathbb{Q}(\alpha)$, complex cubic, one real embedding σ_1 and a conjugate pair σ_2, σ_3 ; unit group $\langle -1, \varepsilon \rangle$ of rank one, regulator $R = \log \sigma_1(\varepsilon) = 2^{O(s)}$ [companion, § 2.3]. For $g \in K$ write $g = c_0(g) + c_1(g)\alpha + c_2(g)\alpha^2$; the index $[O_K : \mathbb{Z}[\alpha]]$ divides $3b$, so $3b c_i(g) \in \mathbb{Z}$ for $g \in O_K$ [companion, § 2.2]. A solution (x, y) of $x^3 - dy^3 = m$ is the element $\beta = x - y\alpha \in \mathbb{Z}[\alpha]$ with $N(\beta) = m$ and **Thue shape** $z(\beta) := c_2(\beta) = 0$. We write $M := \mathbb{Z} + \mathbb{Z}\alpha$ for the rank-2 module in which solutions live.

2.2. Compact representations. A **compact representation** (Thiel format) of $\beta \in O_K$ is a power product

$$\beta = \beta_0 \cdot \prod_{i=1}^T \beta_i^{2^i},$$

with $T = O(\log h(\beta))$ factors, each $\beta_i \in K$ given by $\text{poly}(s)$ bits of coordinate and denominator data. The existence and size bounds are Thiel’s, **unconditional and in arbitrary degree** [Thi95, Def. 6.3.1, Cor. 6.3.9–6.3.10]: every unit and every generator of a principal ideal admits a compact representation with $T \leq \log(\log n + (n-1) \log H) + 2 = O(\log h(\beta))$ factors of $\text{poly}(\log |\Delta|)$ bits each. (Thiel’s normal form uses descending exponents 2^{T-i} and rational factors α_i/d_i , $d_i \leq \sqrt{\Delta}$ — cosmetically, not mathematically, different from the form displayed above; the leading factor additionally carries $O(\log |N(\beta)|)$ bits, harmless here.) Efficient computation from power-product data and log-embedding approximations is Biasse–Fieker [BF14, Alg. 7, Prop. 5.1], unconditional given its input. The quadratic antecedent is [BTW95].

For our instances a solution element has log-height $O(R + \log |m|) = 2^{O(s)}$ (unit reduction and the two-sided window of [companion, Lemmas U and C]), so $T = O(s)$: the certificate has $\text{poly}(s)$ size. Existence is *constructive*: the honest prover runs the companion’s exponential algorithm once; the tracked-generator walk of [companion, Lemma A’], compacted into Thiel’s normal form via [BF14, Alg. 7], emits exactly this format. (Schoof’s Arakelov machinery [Sch08] supplies the reduced-divisor arithmetic used elsewhere but contains no compact-representation theorem — a citation-scope point verified against the source.)

2.3. The certificate is the solution element itself.

Certificate (YES instances). A Thiel-format compact representation of $\beta = x - y\alpha$.

No ε , no reduced representative, no orbit index k , no window: if β is a solution, its compact representation alone witnesses it. In particular the strong window conjecture of [companion, Remark 7.4] is *not needed* for YES-certificates at all; it re-enters only for the prover-efficiency narrative and on the coNP side (Section 6.2).

2.4. The verifier, step by step.

- (1) **V1 — Integrality:** $\beta \in O_K$, and $\beta \in \mathbb{Z}[\alpha]$ after the $3b$ -bookkeeping. Deterministic polynomial time: the format carries denominator data checked by exact ideal arithmetic on poly-size objects; and since coordinate denominators in O_K divide $3b$, membership in $\mathbb{Z}[\alpha]$ is a *congruence* condition modulo $3b$, computable by repeated squaring in a poly(s)-size quotient ring. (*Mechanism sketch; the format-level details are Thiel's.*)
- (2) **V2 — Norm:** $N(\beta) = m$. Deterministic polynomial time, because **norms are multiplicative on power products:**

$$N(\beta) = N(\beta_0) \cdot \prod_{i=1}^T N(\beta_i)^{2^i},$$

each $N(\beta_i)$ a small explicit rational, exponents in binary — exactly the magic that powered Lagarias's Pell certificates [Lag79]. Verifying the exponent identity without factoring is standard multiplicative technology: coprime-basis bookkeeping [vG25], or a special case of Ge's multiplicative-relation test [Ge93]; the sign is read off certified embedding data. (*Sketch; the point is that every ingredient is multiplicative.*)

- (3) **V3 — Thue shape:** $z(\beta) = 0$. **This is the entire remaining difficulty.** It is testable *randomly* in polynomial time (Section 3); whether *deterministically* is precisely CVP (Section 4).

Completeness of the architecture is immediate: $\beta \in \mathbb{Z}[\alpha]$ with $N(\beta) = m$ and $c_2(\beta) = 0$ is a solution $(x, y) = (c_0(\beta), -c_1(\beta))$, and conversely.

3. MERLIN-ARTHUR MEMBERSHIP

Proposition 3.1 (MA membership, unconditional). *Pure cubic Thue solvability is in MA, with perfect completeness and one-sided, amplifiable soundness error.*

Proof (complete modulo the compact-representation format bounds quoted in Section 2.2).

Merlin sends the compact representation of a claimed solution element β . Arthur runs V1 and V2 of Section 2.4 deterministically, then tests V3 modulo random primes.

The integer to test. Since $\alpha, \beta \in O_K$, the trace $W := \text{Tr}_{K/\mathbb{Q}}(\beta\alpha)$ is a rational integer, and by the trace identity of Section 4.2,

$$z(\beta) = 0 \iff W = 0.$$

Height bound from the compact data. Embeddings are multiplicative on the format: for each j ,

$$\log |\sigma_j(\beta)| = \log |\sigma_j(\beta_0)| + \sum_{i=1}^T 2^i \log |\sigma_j(\beta_i)|.$$

Each factor carries $\text{poly}(s)$ bits, so $\log |\sigma_j(\beta_i)| \leq P(s)$ for an explicit polynomial P , and $T = O(s)$ gives $\log_2 \text{house}(\beta) \leq P(s) 2^{T+1} \leq 2^{c_3 s}$ for an explicit c_3 . Hence $|W| \leq 3 \text{house}(\beta) \text{house}(\alpha)$, so $\log_2 |W| \leq 2^{c_3 s} + O(s) \leq 2^{c_4 s}$; if $W \neq 0$, its number of distinct prime divisors obeys $\omega(W) \leq \log_2 |W| \leq 2^{c_4 s}$.

The prime range for a 2/3 threshold. Let E be the set of primes dividing $3d$ or any denominator in the format; $\#E \leq \text{poly}(s)$. Arthur draws p uniformly from the primes in $[17, 2^t]$: by Chebyshev-type bounds ($\pi(x) > x/\ln x$ for $x \geq 17$), the choice $t = c_4 s + O(\log s)$, with explicit absolute constants, supplies at least $3(\omega(W) + \#E)$ primes, so $\Pr[p \in E \text{ or } p \mid W] \leq 1/3$. If $p \in E$, Arthur accepts vacuously (inside the $1/3$). Otherwise he computes $W \bmod p$ in polynomial time: writing multiplication-by- β_i as 3×3 rational matrices M_{β_i} in the power basis, he evaluates $M_\beta \equiv M_{\beta_0} \prod_i M_{\beta_i}^{2^i} \pmod{p}$ by repeated squaring, then $W \equiv \text{tr}(M_\beta M_\alpha) \bmod p$, accepting iff $W \equiv 0$.

Completeness. For a YES instance Merlin sends a true solution element: V1, V2 pass deterministically and $W = 0$, so $W \equiv 0$ modulo *every* prime — Arthur accepts with probability 1.

Soundness (one-sided). For a NO instance, any certificate passing V1 and V2 must have $z(\beta) \neq 0$ — otherwise β would be a solution — hence $W \neq 0$, and Arthur rejects with probability $\geq 2/3$. Arthur never rejects a valid certificate.

Amplification. k independent primes give error $(1/3)^k$; widening the range to 2^{s^c} , $c \geq 2$, already makes a single prime's error exponentially small, since $\omega(W)/\pi(2^{s^c}) \leq 2^{c_4 s - s^c} \text{poly}(s)$. $\square$

Complexity-class bookkeeping. Step V3 alone is a coRP test: vanishing instances are accepted with probability 1, non-vanishing instances rejected with high probability. The outer existential quantifier over certificates therefore places the decision problem in $\exists \cdot \text{coRP} = \text{MA}$ — which is the precise sense of Proposition 3.1.

(Provenance: the notes' ledger listed this as “proposition-with-sketch, needs a careful write-up (height bound on $z(\beta)$ from the compact data; prime-range constants)”. The bounds above discharge those two items; the residual dependence is the quoted format bounds of Section 2.2.)

4. THE COMPACT VANISHING PROBLEM

4.1. Statement and consequence.

CVP_K (Compact Vanishing Problem). Given a compact representation of an algebraic integer β in a fixed cubic field

K , decide in deterministic polynomial time whether a fixed $\mathbb{Q}$ -linear coordinate functional vanishes at β .

For our instances the functional is c_2 , the α^2 -coordinate in $K = \mathbb{Q}(\sqrt[3]{d})$. (The acronym clash with the lattice Closest Vector Problem is noted; no lattice problem appears here.)

Corollary 4.1 (NP iff-modulo CVP). *If $\text{CVP}_K \in \text{P}$ for the instances arising from Thue certificates, then pure cubic Thue solvability is in NP, unconditionally. Conversely — the “iff-modulo” gloss, architectural rather than formal — CVP is the only step of the canonical verifier that is not deterministic polynomial time: any NP proof through this certificate collapses to deciding CVP on these instances. Under standard derandomization hypotheses (which give $\text{MA} = \text{NP}$), NP membership holds conditionally today.*

The notes established three equivalent faces of CVP: a trace form, a lattice-membership form, a bounded-width branching-program form.

4.2. Face one: the trace form. For $\beta = x + y\alpha + z\alpha^2$ one computes $\beta\alpha = zd + x\alpha + y\alpha^2$, and $\text{Tr}(1) = 3$, $\text{Tr}(\alpha) = \text{Tr}(\alpha^2) = 0$ give

$$\text{Tr}(\beta\alpha) = 3dz, \quad \text{i.e.} \quad z(\beta) = \frac{\text{Tr}(\beta\alpha)}{3d}.$$

CVP for our functional is exactly: *is $\text{Tr}(\beta\alpha) = 0$ for compact β ?* The coordinate functionals *are* trace forms — the dual basis of the power basis under the trace pairing gives $c_2(\beta) = \text{Tr}(w\beta)$ with $w = \alpha/(3d)$, and likewise for c_1, c_0 (identities machine-checked during this draft’s assembly).

4.3. Face two: plane membership for Arakelov-pinned elements. The second face rests on a small clean result, the micro-yield of the notes’ Ge-lattice attempt.

Lemma 4.3 (Arakelov pinning). *From a valid compact representation of $\beta \in O_K$, a deterministic polynomial-time verifier computes the full **Arakelov divisor** of β : the exact ideal $I = (\beta)$ in Hermite normal form, and the log-embeddings $\log |\sigma_j(\beta)|$ to any requested poly(s) precision. This data determines β up to sign.*

Proof sketch. The ideal: clear the format’s denominators first and work with the numerator ideal of the compacted element — so that I is integral, as the HNF intersection $L = I \cap M$ of Section 4.3 strictly requires — then reduce the power product by repeated squaring on reduced ideals with tracked distances — infrastructure arithmetic is precisely this machinery [BW88, Sch08] — every intermediate object staying poly-size. The logs: interval arithmetic on the factors’ certified embedding data, via the exact telescoping of Section 3. Uniqueness: if $(\beta) = (\beta')$ with equal log vectors, β/β' is a unit with vanishing log-embeddings, hence torsion, hence ± 1 (the torsion of K is $\{\pm 1\}$ by the real embedding). (*sketch*) $\square$

So the certificate pins a *unique* candidate (up to sign, harmless since $z(-\beta) = -z(\beta)$): **compact certificates verify everything about β except one plane membership**. Indeed $z(\beta) = 0 \iff \beta \in M$, and $L := I \cap M$ is an explicit rank-2 lattice with a poly-size basis (HNF intersection). Hence:

**CVP (Thue instances) $\iff$ does the element pinned
by a given Arakelov divisor lie in an explicit rank-2
sublattice?**

4.4. Face three: width-3 ABP trace zero-testing. Writing multiplication-by- β_i as 3×3 integer matrices (after clearing the format's denominators), the compact representation becomes a **matrix word with repeated-squaring structure**, and by the trace form CVP reads: *decide whether the trace of a width-3 integer matrix word is zero*. That is polynomial identity testing for width-3 algebraic branching programs — a highly structured, white-box, repeated-squaring instance, and the right literature door: it connects the NP question for cubic Thue equations to bounded-width PIT derandomization by the shortest path we know of (Section 5.4). One structural advantage must be stated explicitly: the matrices M_{β_i} all lie in the commutative subalgebra $\mathbb{Q}[M_\alpha] \cong K$ of 3×3 matrices — the word is not a generic non-commutative width-3 ABP but one over a **commutative** matrix ring, a class significantly better understood in the PIT literature and correspondingly a more tempting target for derandomization.

4.5. Remark (Lagarias vacuity: why 1979 stopped at degree 2).

Remark 4.5. In the quadratic case the module $\mathbb{Z} + \mathbb{Z}\sqrt{d}$ is *all* of $\mathbb{Z}[\sqrt{d}]$: the Thue-shape condition is vacuous, CVP never arises, and norm-checking alone suffices — which is why Lagarias's certificates [Lag79] close at degree 2 with purely multiplicative technology. The coordinate condition is genuinely **new at degree 3**: CVP is exactly what separates the cubic certificate problem from Pell — this, not the unit computation, is the honest frontier of the NP question, and the explanation of the shift from degree 2 (NP since 1979) to degree 3 (MA now).

5. WHY THE CLASSICAL TOOLBOX STOPS

We place CVP on the map, then close four attack routes — distilling the notes' 1991–2026 literature sweep and recorded attempts. No route is closed by a hardness theorem; each is closed by structural inapplicability, a different and more instructive thing.

5.0. Where CVP sits: the sum-of-square-roots family. The cleared trace value admits a $\text{poly}(s)$ -size straight-line program (repeated squaring), so CVP is an instance of EquSLP — zero-testing SLP-given integers — in coRP (the random-prime test of Section 3; the observation goes back to Schönhage [Sch79]), with its positivity cousin PosSLP in the counting

hierarchy [ABKM09, ESY14]. Deterministic polynomial time is open for the whole family, whose totem is the sum-of-square-roots problem: poly-time for explicitly given radical sums [Blo91], uniform TC^0 for deciding whether a rational radical sum is rational [HBMOW10], the separation-bound frontier only recently moved by the subspace theorem [EHS24]. The notes' verdict, adopted here: CVP is **genuinely new, of sum-of-square-roots-class flavor** — in coRP, in CH, deterministic case open — with one unexplored asset recorded in Section 7.2.

5.1. The additive/multiplicative divide. Thiel-style deterministic *equality* testing of compact representations is a **multiplicative** technology: $\beta = \gamma$ reduces to the quotient being 1, and units $\neq \pm 1$ are separated from 1 by Dobrowolski-type height lower bounds [Dob79] on the log-embedding lattice — polynomial precision suffices. Ge's theorem [Ge93] (see [vG25, Thm. 165]) extends this: *multiplicative relations* among power products are decidable in deterministic polynomial time — exactly why the norm step V2 is easy. The compact-representation toolbox stops at the trace, precisely: [BTW95] adds two-term sign comparison in real quadratic fields, and **no additive ≥ 3 -term test exists anywhere in the literature.**

CVP is **additive**: $\text{Tr}(\beta\alpha)$ is a sum of three conjugate terms. Separation is not the problem — integrality gives it for free (the cleared value is a rational integer, so nonzero means ≥ 1). **Evaluation is the problem**: certifying the sum numerically needs absolute precision comparable to the archimedean spread $\max_j \log |\sigma_j(\beta)| - \min_j \log |\sigma_j(\beta)|$, and for *true* Thue solutions the spread is $2^{\Theta(s)}$ forced: a solution has $|\sigma_1(\beta)| = |m|/|\sigma_2(\beta)|^2$ tiny *precisely because* it is a solution. The easy “balanced- β ” case of CVP can therefore never contain the instances we care about. The angle formulation hits the same wall: verifying $\arg \sigma_2(\beta\alpha) \approx \pm\pi/2$ to error $e^{-\text{spread}}$ needs the small factors' arguments to exponentially many bits.

The conjugate-certificate attempt, and its circle. One can move to the Galois closure $L = K(\omega)$ and note $\text{Tr}(\beta\alpha) = 0 \iff \beta + \omega\sigma_2(\beta) + \omega^2\sigma_3(\beta) = 0$; certifying compact representations of the conjugates is fine, but the final check is again an additive vanishing of compact objects — the divide reappears intact. **Sums of compact representations are not compact; that sentence is the obstacle.**

5.2. Blömer is structurally inapplicable, not merely insufficient. Blömer's deterministic polynomial-time algorithms for sums of radicals [Blo91, Blo93] might look like the natural import: our elements are $\mathbb{Q}$ -combinations of $1, d^{1/3}, d^{2/3}$. But his determinism rides on *Siegel-type rigidity*: independent radicals admit **no** nontrivial $\mathbb{Q}$ -linear relations, so zero-testing reduces to coefficient collection on explicit poly-size coefficients. Our trace-zero instances **are** nontrivial relations in the rank-2 trace-zero module — the exact phenomenon his theorem excludes — with coefficients x, y, z compact rather than explicit. Both pillars — rigidity and explicit arithmetic — fail simultaneously.

5.3. No ideal-theoretic route: the hyperplane argument. The tempting program “tear into Thiel’s ideal equality tests to break CVP” is closed before it opens, by rank counting.

Proposition 5.1. *No ideal containment, equality, or divisibility statement in O_K can express the condition $z(\beta) = 0$, even in principle.*

Proof. The trace-zero condition $\text{Tr}(\beta\alpha) = 0$ defines a $\mathbb{Q}$ -hyperplane in K (the coordinate functionals are trace forms, Section 4.2). Ideals are **full-rank** $\mathbb{Z}$ -modules: any nonzero ideal of O_K has rank 3, while any module contained in the hyperplane has rank ≤ 2 . $\square$

Thiel/Ge machinery tests relations among full-rank multiplicative objects; CVP lives in a corank-1 additive slice invisible to all of them. The bridge of Section 5.5 is not hiding inside ideal arithmetic; it must be genuinely new.

5.4. The EquSLP caution, and the Baker route’s ceiling. Two doors from complexity theory, each honestly measured:

- **Iterated squaring is the hard core.** ABKM [ABKM09, Prop. 2.1] show that EquSLP’s difficulty concentrates on repeated-squaring instances — our shape — and derandomizing such identity tests implies circuit lower bounds [KI04]. This is a **caution, not a hardness proof**, for our width-3 single words: CVP could be easy without lower-bound consequences, but the precedent says the general-toolbox route is priced.
- **The Baker route has a height ceiling.** Galby–Ouaknine–Worrell [GOW15] decide sign and zero questions for entries of M^n , $\dim \leq 3$, in polynomial time — for **poly-height** inputs: the engine is lower bounds for linear forms in logarithms [Mat00], whose constants stay polynomial only while heights and the number of logarithms stay bounded. Thue certificates have $h(\varepsilon) = R/3 = 2^{\Theta(s)}$; at $\Theta(s)$ logarithms the Baker constants degrade exponentially — the same wall that pins LRS Positivity at order ≥ 6 [OW14]. The bounded fragment this route *does* give is Proposition 6.1.

5.5. One wall, four costumes; the missing tool, named. The four failures above are one wall:

- (1) **Trace non-multiplicativity** (Section 5.1).
- (2) **Sums of compacts are not compact** (Section 5.1).
- (3) **The archimedean spread** (Section 5.1): $2^{\Theta(s)}$ precision forced exactly on true solutions.
- (4) **Plane non-stability** (the notes’ Ge-lattice attempt): the rescaling trick — bring the thin strip $\{|x - y\alpha| \sim e^{\ell_1}\}$ to bounded size by ε^{-j} — dies because M is not stable under units: $\varepsilon^{-j}M \neq M$. Lattice-point location in the plane is polynomial *at polynomial scales*, but our scales are $e^{2^{O(s)}}$, described only by their logs.

What a solution needs, stated once: a **multiplicative-to-additive bridge** — any theorem letting membership in a fixed rank-2 plane be tested through data carried multiplicatively (relation lattices, Arakelov classes, unit orbits). Nothing in the 1991–2026 literature provides one; building it is the actual open problem, and by Section 4.3 it now has a precise geometric formulation.

6. POSITIVE FRAGMENTS

6.1. The bounded fragment: CVP at bounded rank and small height.

Proposition 6.1 (bounded CVP — stated with a proof plan). *Restrict CVP_K to instances whose factor list $\{\beta_0, \dots, \beta_T\}$ generates a subgroup of $K^\times/\{\pm 1\}$ of multiplicative rank $O(1)$ admitting generators of Weil height $\text{poly}(s)$. This fragment is decidable in deterministic polynomial time.*

Proof plan (honestly flagged: a plan, not a proof). (1) Compute the relation lattice of the factors — deterministic polynomial time by Ge [Ge93], [vG25, Thm. 165] — and rewrite the word over $O(1)$ independent poly-height generators with binary exponents. (2) The trace becomes a fixed short sum of conjugate power products in $O(1)$ poly-height bases. (3) Zero-test via the Baker–Matveev machinery as in the dimension-3 matrix-power analysis of [GOW15]: with $O(1)$ logarithms of poly-height numbers and binary exponents, Matveev’s bounds [Mat00] are polynomial — noting that Matveev’s constant degrades exponentially in the number k of logarithms, which is exactly why the $O(1)$ multiplicative-rank restriction is load-bearing and cannot be relaxed, so certified interval evaluation at $\text{poly}(s)$ precision decides against the free integrality separation. Steps (2)–(3) need the same case-discipline GOW exercise at dimension 3 and are not written out here. $\square$

What this fragment is and is not. It is the notes’ verdict “publishable-but-bounded”. It does **not** cover Thue certificates: their factor lists contain ε (or its walk factors), of height $R/3 = 2^{\Theta(s)}$ — precisely the ceiling of Section 5.4. No claim of progress on the Thue NP question is made here.

6.2. The strong window conjecture: prover efficiency and coNP.

The strong window conjecture [companion, Remark 7.4] — every Thue-shape index of a unit-reduced representative satisfies $|k| \leq C(1 + \log |m|/R)$, hence $O(s)$ by the regulator floor [ADF16] — turned out to be unnecessary for YES-certificates (Section 2.3). Its remaining roles: **prover efficiency** — the honest prover runs the companion’s algorithm once ($2^{O(s)}$); the conjecture caps the final scan at $O(s)$ indices per representative, though the infrastructure walk remains $\tilde{\Theta}(R)$, so it trims the narrative, not the exponential — and **the coNP side**, where certifying a NO instance would require certifying every ideal class’s orbit window zero-free, and the conjecture caps the indices to certify per orbit at $\text{poly}(s)$, making class-enumeration NO-certificates conceivable. (*Ledger flag, preserved: the coNP side is untouched in this work.*)

7. OPEN PROBLEMS

7.1. CVP itself. Is CVP_K — even just for Thue-shaped instances — in P? The notes’ two probed routes are blocked, not dead: the *p-adic route* (ord_p of power products is poly-computable per prime) stumbles on the fact that valuations do not see addition; the *canonical-form route* (Thiel equality of β against its projection) needs a compact representation of the projection, and sums of compacts are not compact. The sharpest open formulation is the ABP face: **is the repeated-squaring width-3 matrix-word class among the deterministically testable bounded-width PIT cases?** A positive answer completes: cubic Thue $\in \text{NP}$, unconditionally — Lagarias one rung up, for real.

7.2. The multiplicative-to-additive bridge. Build any theorem letting membership in a fixed rank-2 plane be tested through multiplicatively carried data (Section 5.5). The one unexplored asset, recorded in the notes’ literature verdict: our three summands are Galois-conjugate, and the bases’ full multiplicative relation lattice is *computable* (Ge). The additive analog of [vG25, Lemma 166] — detecting trace-zero from the relation lattice plus unit-lattice geometry — has never been attempted. That is the door.

7.3. DIVIS hardness. The program’s synthesis records that the divisibility language $\text{DIVIS} = \{(A, B) \in \mathbb{Z}[x]^2 : \exists x \in \mathbb{Z}, A(x) \mid B(x)\}$ is in NP (two proofs, one the expedition’s, by an elementary integer-remainder argument — Runge in miniature); its hardness is unstudied. It is the nearest candidate for a hardness foothold adjacent to the Thue stratum.

7.4. The coNP certificate question. Is pure cubic Thue solvability in coNP? Degree 2 says yes one rung down (negative Pell is in $\text{NP} \cap \text{coNP}$ [Lag79]). At degree 3 the ingredients on the table are the strong window conjecture and class-enumeration certificates (Section 6.2); nothing beyond the framing exists yet.

Status ledger (carried from the notes, unchanged in spirit).

- MA membership: proposition with the height-bound and prime-range write-up supplied (Section 3); the format-bound dependence is now DISCHARGED by the pinned citations (Section 2.2).
- Compact-representation existence and size: **debt paid** — Thiel’s thesis read in full (Def. 6.3.1, Cor. 6.3.9–6.3.10, unconditional, arbitrary degree), efficient computation pinned to [BF14, Alg. 7, Prop. 5.1]; the erroneous compaction citation to [Sch08] corrected (it contains no compact representations — verified against the source). The remaining ledger flags may now be stripped at final submission.
- CVP: open, named, isolated — the right-sized next mathematics.
- coNP side: untouched here.

REFERENCES

- [ABKM09] E. Allender, P. Bürgisser, J. Kjeldgaard-Pedersen, P. B. Miltersen, *On the complexity of numerical analysis*, SIAM J. Comput. 38:5 (2009) 1987–2006.
- [ADF16] S. Astudillo, F. Díaz y Díaz, E. Friedman, *Sharp lower bounds for regulators of small-degree number fields*, J. Number Theory 167 (2016) 232–258.
- [BF14] J.-F. Biasse, C. Fieker, *Subexponential class group and unit group computation in large degree number fields*, LMS J. Comput. Math. 17 (2014) 385–403. (§ 5, Algorithm 7, Proposition 5.1: efficient computation of compact representations; unconditional given its input.)
- [Blo91] J. Blömer, *Computing sums of radicals in polynomial time*, Proc. 32nd IEEE FOCS (1991) 670–677.
- [Blo93] J. Blömer, *Simplifying expressions involving radicals*, PhD thesis, Freie Universität Berlin, 1993.
- [BTW95] J. Buchmann, C. Thiel, H. C. Williams, *Short representation of quadratic integers*, in: Computational Algebra and Number Theory, Math. Appl. 325, Kluwer, 1995, 159–185.
- [BW88] J. Buchmann, H. C. Williams, *On the infrastructure of the principal ideal class of an algebraic number field of unit rank one*, Math. Comp. 50 (1988) 569–579.
- [companion] The diophantine-fable expedition, *Pure cubic Thue equations are solvable in deterministic exponential time*, companion manuscript, `papers/cubic-thue-exp.md` in this repository (draft of 2026-07-22).
- [Dob79] E. Dobrowolski, *On a question of Lehmer and the number of irreducible factors of a polynomial*, Acta Arith. 34 (1979) 391–401.
- [EHS24] F. Eisenbrand, M. Haeberle, N. Singer, *An improved bound on sums of square roots via the subspace theorem*, Proc. SoCG 2024, LIPIcs 293, art. 54.
- [ESY14] K. Etessami, A. Stewart, M. Yannakakis, *A note on the complexity of comparing succinctly represented integers, with an application to maximum probability parsing*, ACM Trans. Comput. Theory 6:2 (2014), art. 9.
- [Ge93] G. Ge, *Testing equalities of multiplicative representations in polynomial time*, Proc. 34th IEEE FOCS (1993) 422–426; and PhD thesis (*Algorithms related to multiplicative representations of algebraic numbers*), U.C. Berkeley, 1993.
- [GOW15] E. Galby, J. Ouaknine, J. Worrell, *On matrix powering in low dimensions*, Proc. STACS 2015, LIPIcs 30, 329–340.
- [Hal07] S. Hallgren, *Polynomial-time quantum algorithms for Pell’s equation and the principal ideal problem*, J. ACM 54:1 (2007), art. 4.
- [HBMOW10] P. Hunter, P. Bouyer, N. Markey, J. Ouaknine, J. Worrell, *Computing rational radical sums in uniform TC^0* , Proc. FSTTCS 2010, LIPIcs 8, 308–316.
- [KI04] V. Kabanets, R. Impagliazzo, *Derandomizing polynomial identity tests means proving circuit lower bounds*, Comput. Complexity 13 (2004) 1–46.
- [Lag79] J. C. Lagarias, *Succinct certificates for the solvability of binary quadratic Diophantine equations*, Proc. 20th IEEE FOCS (1979) 47–54; extended version arXiv:math/0611209.
- [Mat00] E. M. Matveev, *An explicit lower bound for a homogeneous rational linear form in the logarithms of algebraic numbers. II*, Izv. Math. 64:6 (2000) 1217–1269.
- [OW14] J. Ouaknine, J. Worrell, *Positivity problems for low-order linear recurrence sequences*, Proc. SODA 2014, 366–379.
- [Sch79] A. Schönhage, *On the power of random access machines*, Proc. ICALP 1979, LNCS 71, 520–529.
- [Sch08] R. Schoof, *Computing Arakelov class groups*, in: Algorithmic Number Theory, MSRI Publ. 44, Cambridge University Press, 2008, 447–495.

- [Sha19] M. Sha, *Effective results on the Skolem Problem for linear recurrence sequences*, J. Number Theory 197 (2019) 228–249.
- [Sma98] S. Smale, *Mathematical problems for the next century*, Math. Intelligencer 20:2 (1998) 7–15.
- [Thi95] C. Thiel, *On the complexity of some problems in algorithmic algebraic number theory*, PhD thesis, Universität des Saarlandes, Saarbrücken, 1995 (Definition 6.3.1, Corollaries 6.3.9–6.3.10; Theorem 6.4.1: principal-ideal certificates in NP, unconditional); journal companion: *Short proofs using compact representations of algebraic integers*, J. Complexity 11 (1995) 310–329; announcement: ANTS-I 1994, LNCS 877, 234–247.
- [vG25] D. M. H. van Gent (lectures by H. W. Lenstra), *Polynomial-time algorithms in algebraic number theory*, arXiv:2502.19036. (Thm. 165 = Ge’s theorem; Lemma 166; coprime bases.)